Network Project Verification & Testing Report

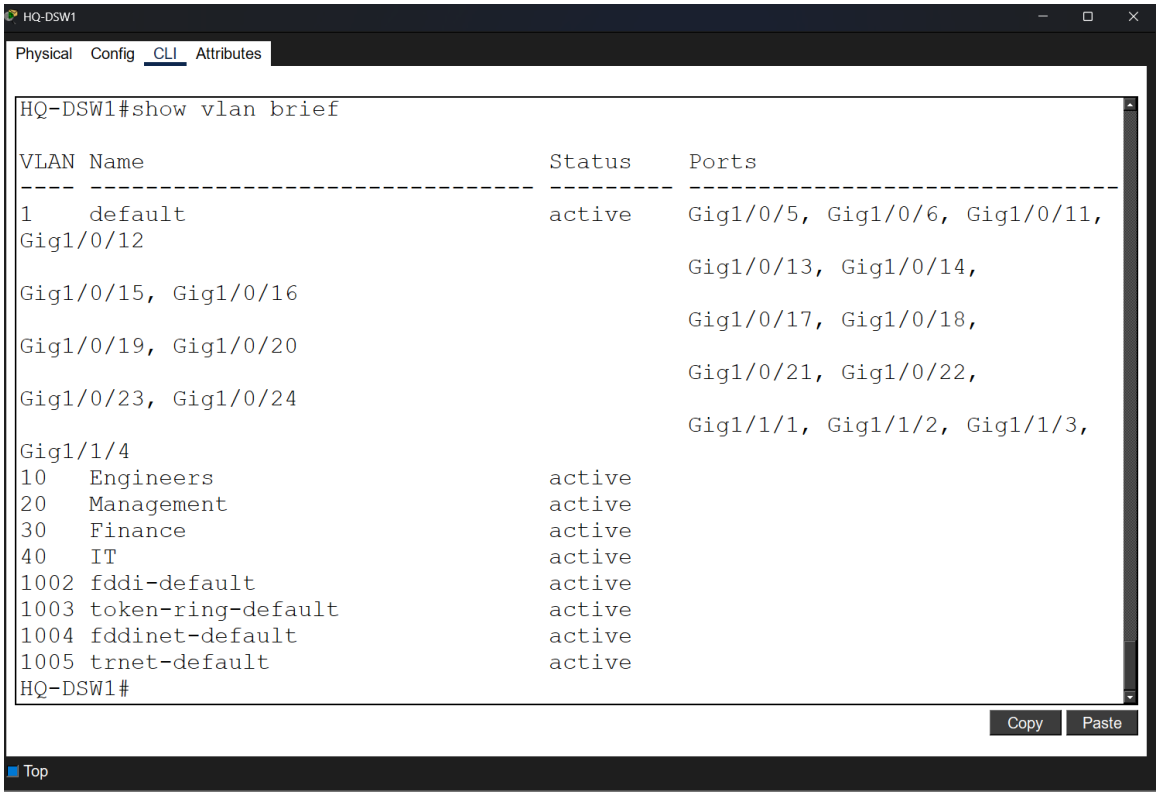
1. VLAN Configuration Verification

Description:

This section verifies that VLANs are correctly created on the distribution switches at both HQ and BR sites. The screenshots below show the VLAN IDs and names configured on each switch.

Screenshot:

This screenshot shows the VLAN configuration on HQ-DSW1. VLANs 10 (Engineers), 20 (Management), 30 (Finance), and 40 (IT) are successfully created as required.



This screenshot shows the VLAN configuration on BR-DSW1. VLANs 60 (Maintenance), 70 (HR), and 80 (Sales) are successfully created according to the project requirements.

```
BR-DSW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10
                                                Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
                                                Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18
                                                Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22
                                                Gig1/0/23, Gig1/0/24, Gig1/1/1, Gig1/1/2
                                                Gig1/1/3, Gig1/1/4
60   Maintenance                      active
70   HR                               active
80   Sales                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
BR-DSW1#
```

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2. Trunk Links Verification

Description:

This section verifies that trunk links are properly configured between switches to allow VLAN traffic to pass across the network. Trunk ports ensure that multiple VLANs can communicate between distribution switches.

Screenshot:

This screenshot shows the trunk configuration on HQ-DSW1. The trunk ports are active and allow multiple VLANs to pass through the switch.

HQ-DSW1

Physical Config CLI Attributes

```
HQ-DSW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/7    on           802.1q         trunking      1
Gig1/0/8    on           802.1q         trunking      1
Gig1/0/9    on           802.1q         trunking      1
Gig1/0/10   on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/7    1-1005
Gig1/0/8    1-1005
Gig1/0/9    1-1005
Gig1/0/10   1-1005

Port        Vlans allowed and active in management domain
Po1         1,10,20,30,40
Gig1/0/7    1,10,20,30,40
Gig1/0/8    1,10,20,30,40
Gig1/0/9    1,10,20,30,40
Gig1/0/10   1,10,20,30,40
```

This screenshot shows the trunk configuration on HQ-DSW2. The trunk links are correctly established and operational.

```
HQ-DSW2
Physical  Config  CLI  Attributes

HQ-DSW2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/6    auto         n-802.1q       trunking      1
Gig1/0/7    on           802.1q         trunking      1
Gig1/0/8    on           802.1q         trunking      1
Gig1/0/9    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/6    1-1005
Gig1/0/7    1-1005
Gig1/0/8    1-1005
Gig1/0/9    1-1005

Port        Vlans allowed and active in management domain
Po1         1,10,20,30,40
Gig1/0/6    1,10,20,30,40
Gig1/0/7    1,10,20,30,40
Gig1/0/8    1,10,20,30,40
Gig1/0/9    1,10,20,30,40
```

This screenshot shows the trunk configuration on BR-DSW1. The trunk ports are successfully configured to carry VLAN traffic.

```
BR-DSW1
Physical  Config  CLI  Attributes

BR-DSW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1
Gig1/0/4    on           802.1q         trunking      1
Gig1/0/5    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/3    1-1005
Gig1/0/4    1-1005
Gig1/0/5    1-1005

Port        Vlans allowed and active in management domain
Po1         1,60,70,80
Gig1/0/3    1,60,70,80
Gig1/0/4    1,60,70,80
Gig1/0/5    1,60,70,80

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,60,70,80
Gig1/0/3    1,60,70,80
Gig1/0/4    1,60,70,80
```

This screenshot shows the trunk configuration on BR-DSW2. The trunk links are active and functioning properly between switches.

```
BR-DWS2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1
Gig1/0/4    on           802.1q         trunking      1
Gig1/0/5    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/3    1-1005
Gig1/0/4    1-1005
Gig1/0/5    1-1005

Port        Vlans allowed and active in management domain
Po1         1,60,70,80
Gig1/0/3    1,60,70,80
Gig1/0/4    1,60,70,80
Gig1/0/5    1,60,70,80

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,60,70,80
Gig1/0/3    1,60,70,80
Gig1/0/4    1,60,70,80
```

3. EtherChannel Verification

Description:

This section verifies that EtherChannel is correctly configured between distribution switches to provide higher bandwidth and redundancy. Multiple physical links are combined into a single logical Port-Channel using LACP in HQ and PAgP in BR.

Screenshot:

This screenshot shows the EtherChannel configuration on HQ-DSW1. The Port-Channel is successfully formed using LACP and is in an active state.

```
HQ-DSW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/2(P) Gig1/0/3(P) Gig1/0/4(P)
HQ-DSW1#
```

This screenshot shows the EtherChannel configuration on HQ-DSW2. The Port-Channel is correctly established and functioning using LACP.

```
HQ-DSW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/2(P) Gig1/0/3(P) Gig1/0/4(P)
HQ-DSW2#
```

This screenshot shows the EtherChannel configuration on BR-DSW1. The Port-Channel is successfully configured using PAgP and is operational.

```
BR-DSW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Gig1/0/1(P) Gig1/0/2(P)
BR-DSW1#
```

This screenshot shows the EtherChannel configuration on BR-DSW2. The Port-Channel is properly formed using PAgP and working as expected.

```
BR-DWS2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Gig1/0/1(P) Gig1/0/2(P)
BR-DWS2#
```

4. VTP Configuration Verification

Description:

This section verifies that VLAN Trunking Protocol (VTP) is correctly configured across all distribution switches. VTP ensures VLAN consistency by propagating VLAN information within the same domain. All switches are configured with the same VTP domain name, version, and password.

Screenshot:

This screenshot shows the VTP configuration on HQ-DSW1. The switch is configured in server mode with the correct domain name "Nass.com", version 2, and matching VTP password.

```
HQ-DSW1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : Nass.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.1677.C750
Configuration last modified by 0.0.0.0 at 3-25-26 05:58:37
Local updater ID is 192.168.10.1 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 9
Configuration Revision            : 201
MD5 digest                        : 0xAC 0xAF 0xBB 0xF5 0xD5 0x33 0xD1 0x0F
                                    0x70 0x5E 0x60 0x05 0x72 0x2D 0xF7 0x05
HQ-DSW1#
```

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This screenshot shows the VTP configuration on HQ-DSW2. The switch is operating in client mode and successfully receives VLAN information from the VTP server.

```
HQ-DSW2#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : Nass.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.6326.DDA0
Configuration last modified by 0.0.0.0 at 3-25-26 05:58:37

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 9
Configuration Revision            : 201
MD5 digest                        : 0xAC 0xAF 0xBB 0xF5 0xD5 0x33 0xD1 0x0F
                                    0x70 0x5E 0x60 0x05 0x72 0x2D 0xF7 0x05
HQ-DSW2#
```

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This screenshot shows the VTP configuration on BR-DSW1. The switch is configured in server mode with the correct VTP domain and settings.

```
BR-DSW1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : Nass.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 000A.F37D.6800
Configuration last modified by 0.0.0.0 at 3-25-26 05:58:37
Local updater ID is 192.168.60.1 on interface Vl60 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 8
Configuration Revision            : 152
MD5 digest                        : 0x3B 0x63 0xE9 0xD8 0x7F 0x5B 0xE9 0x34
                                    0x66 0x73 0xAE 0x38 0xAC 0x0C 0x90 0xAD
BR-DSW1#
```

This screenshot shows the VTP configuration on BR-DSW2. The switch is in client mode and correctly synchronized with the VTP server.

```
BR-DWS2#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : Nass.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.4ADC.7800
Configuration last modified by 0.0.0.0 at 3-25-26 05:58:37

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 8
Configuration Revision            : 152
MD5 digest                        : 0x3B 0x63 0xE9 0xD8 0x7F 0x5B 0xE9 0x34
                                    0x66 0x73 0xAE 0x38 0xAC 0x0C 0x90 0xAD
BR-DWS2#
```

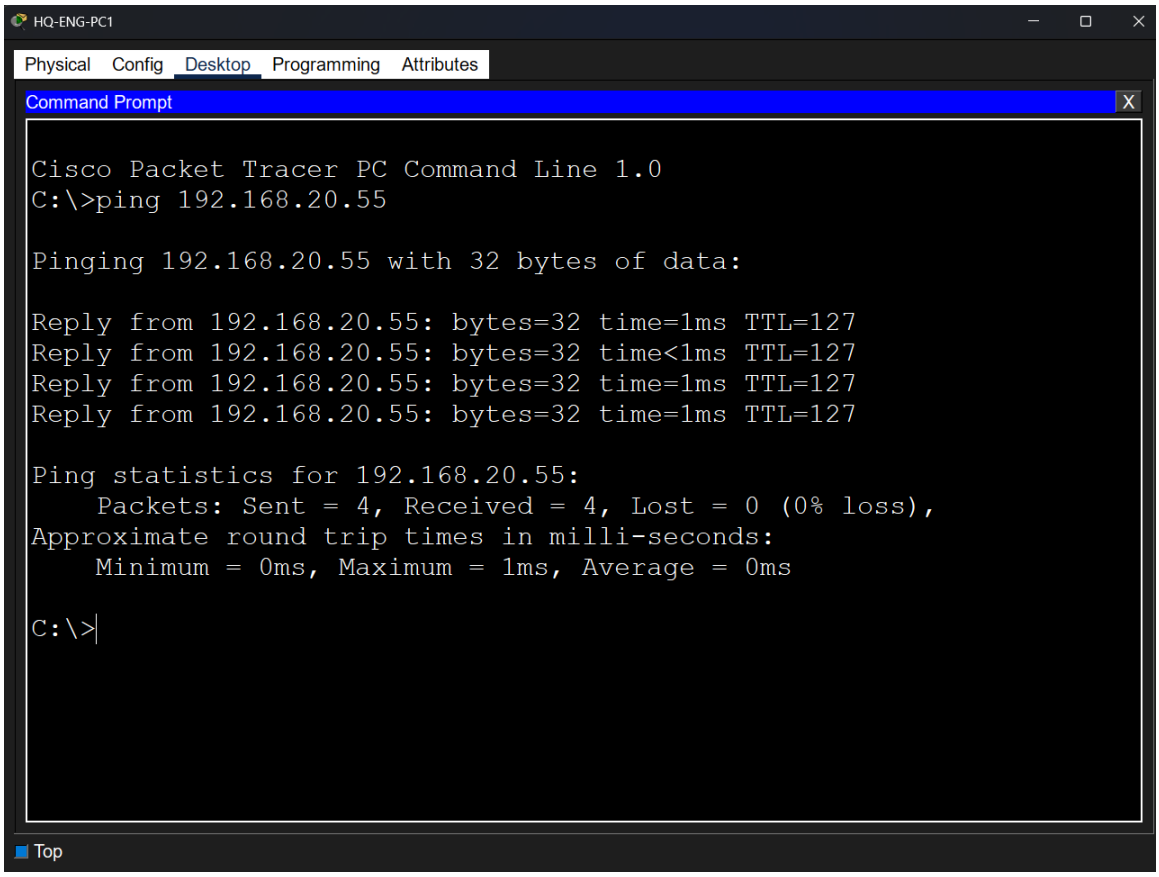
5. Inter-VLAN Connectivity (Ping Test)

Description:

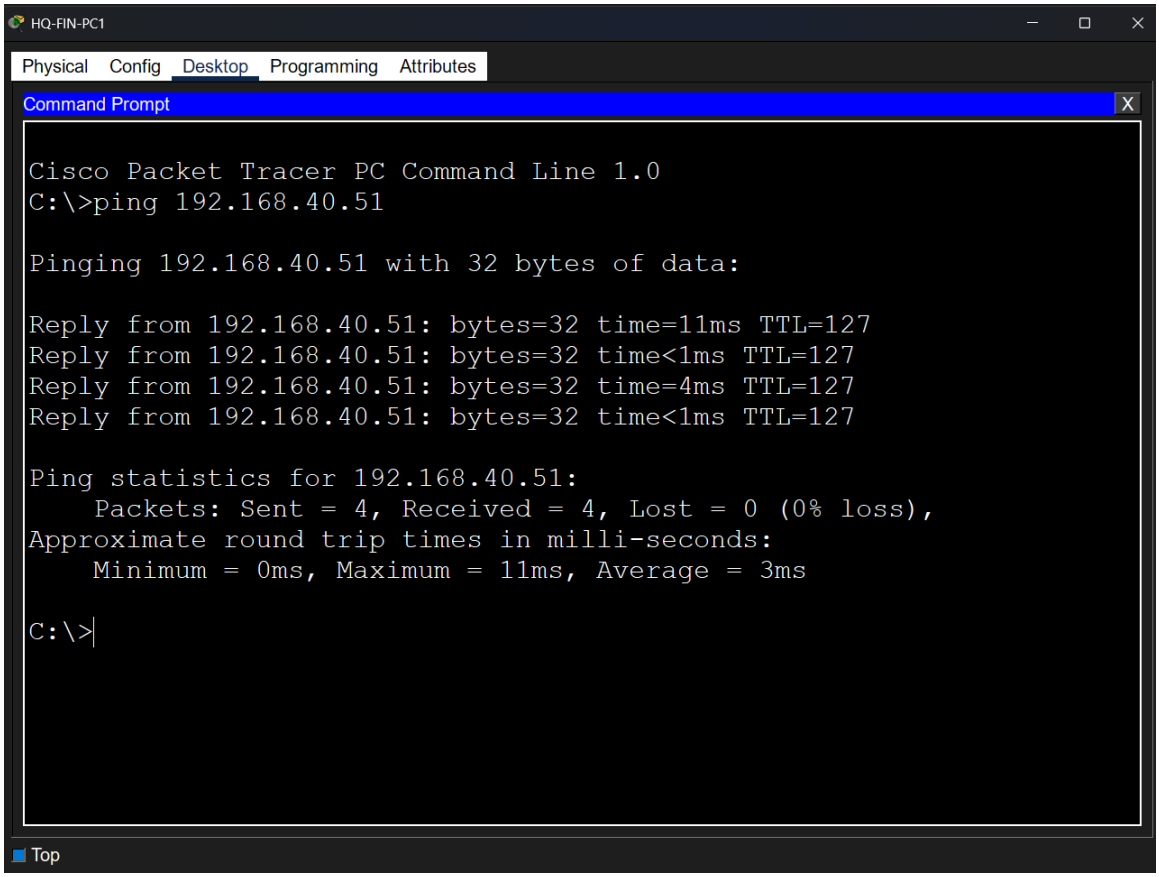
This section verifies successful communication between different VLANs within the same site using Inter-VLAN routing. The ping results confirm that devices in separate VLANs can communicate through the Layer 3 distribution switches.

Screenshot:

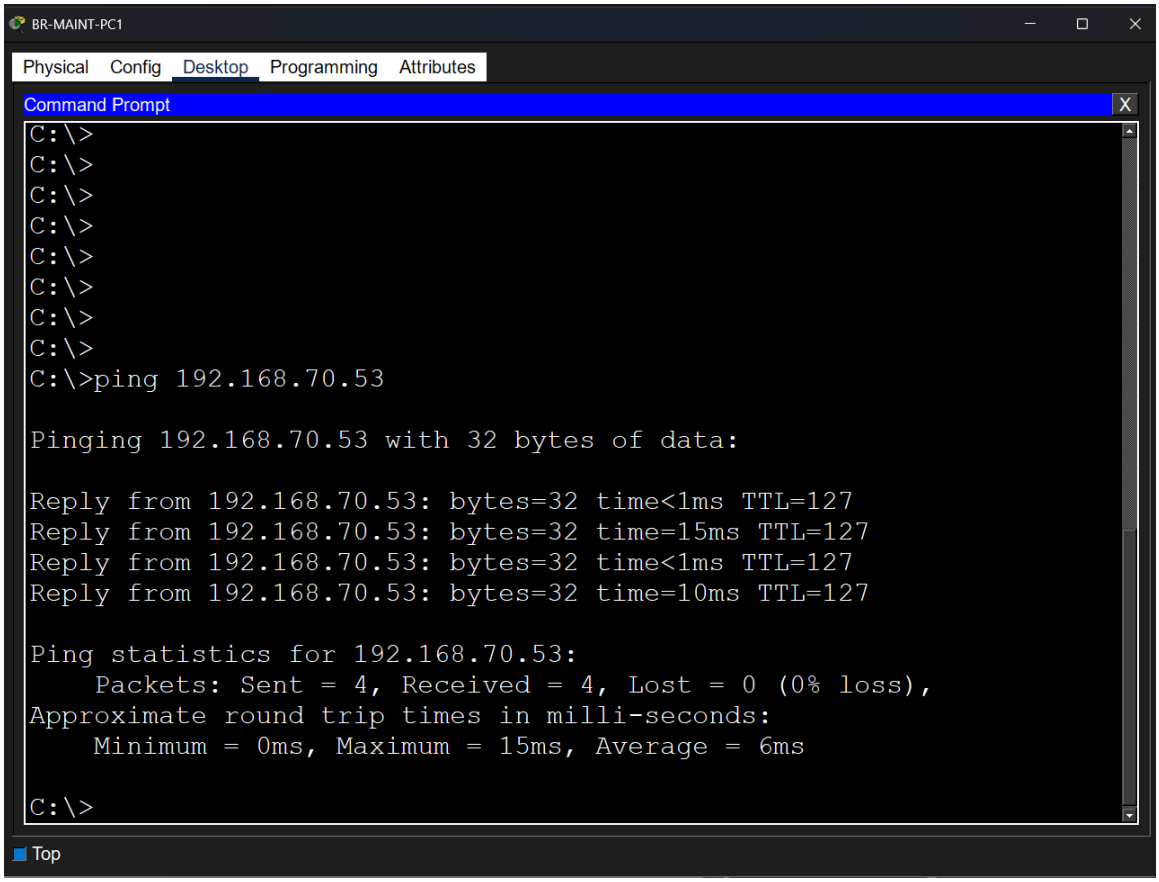
This screenshot shows a successful ping from a PC in VLAN 10 to a device in VLAN 20 (192.168.20.55). The replies confirm proper inter-VLAN routing.


HQ-ENG-PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.55
Pinging 192.168.20.55 with 32 bytes of data:
Reply from 192.168.20.55: bytes=32 time=1ms TTL=127
Reply from 192.168.20.55: bytes=32 time<1ms TTL=127
Reply from 192.168.20.55: bytes=32 time=1ms TTL=127
Reply from 192.168.20.55: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.20.55:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

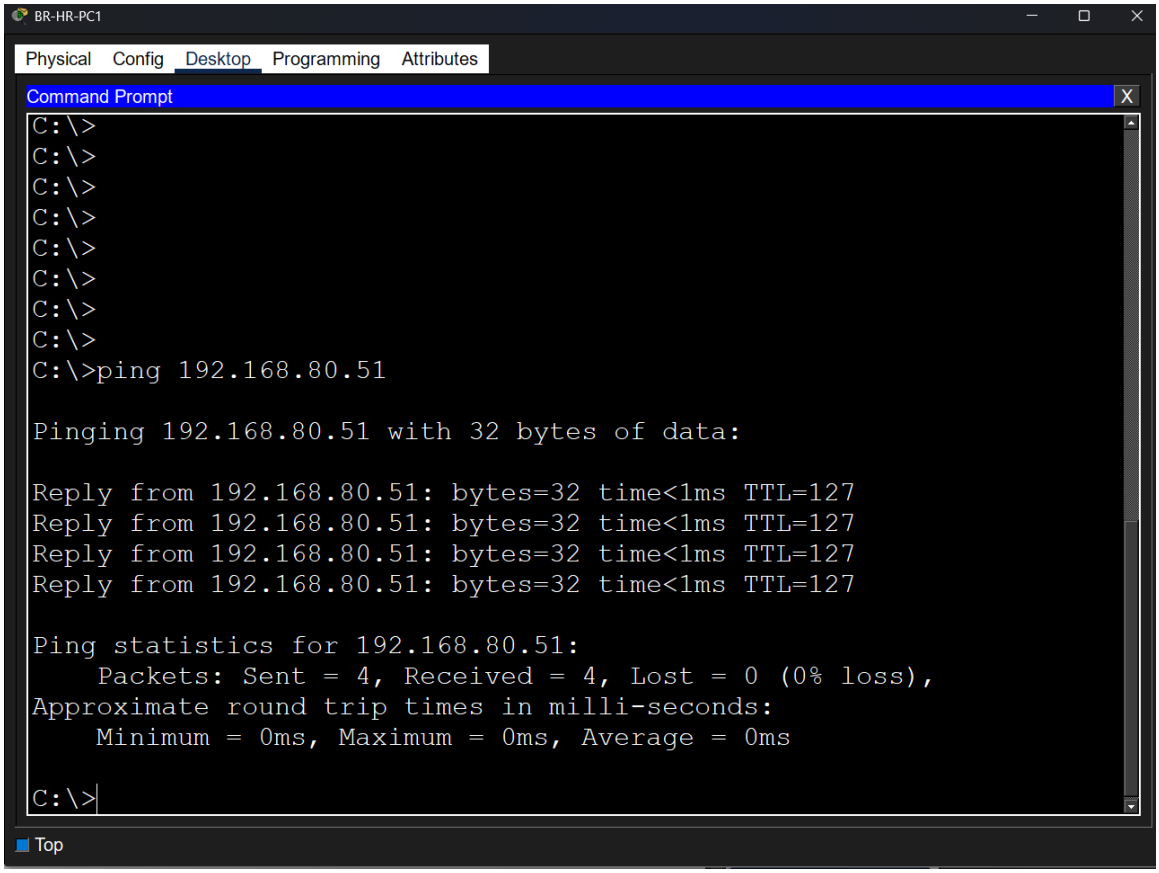
This screenshot shows a successful ping from a PC in VLAN 30 to a device in VLAN 40 (192.168.40.51). This verifies connectivity between these VLANs.



This screenshot shows a successful ping from a device in VLAN 60 to a PC in VLAN 70 (192.168.70.53). The communication is working correctly.



This screenshot shows a successful ping from a device in VLAN 70 to a device in VLAN 80 (192.168.80.51). This confirms inter-VLAN communication in the BR site.



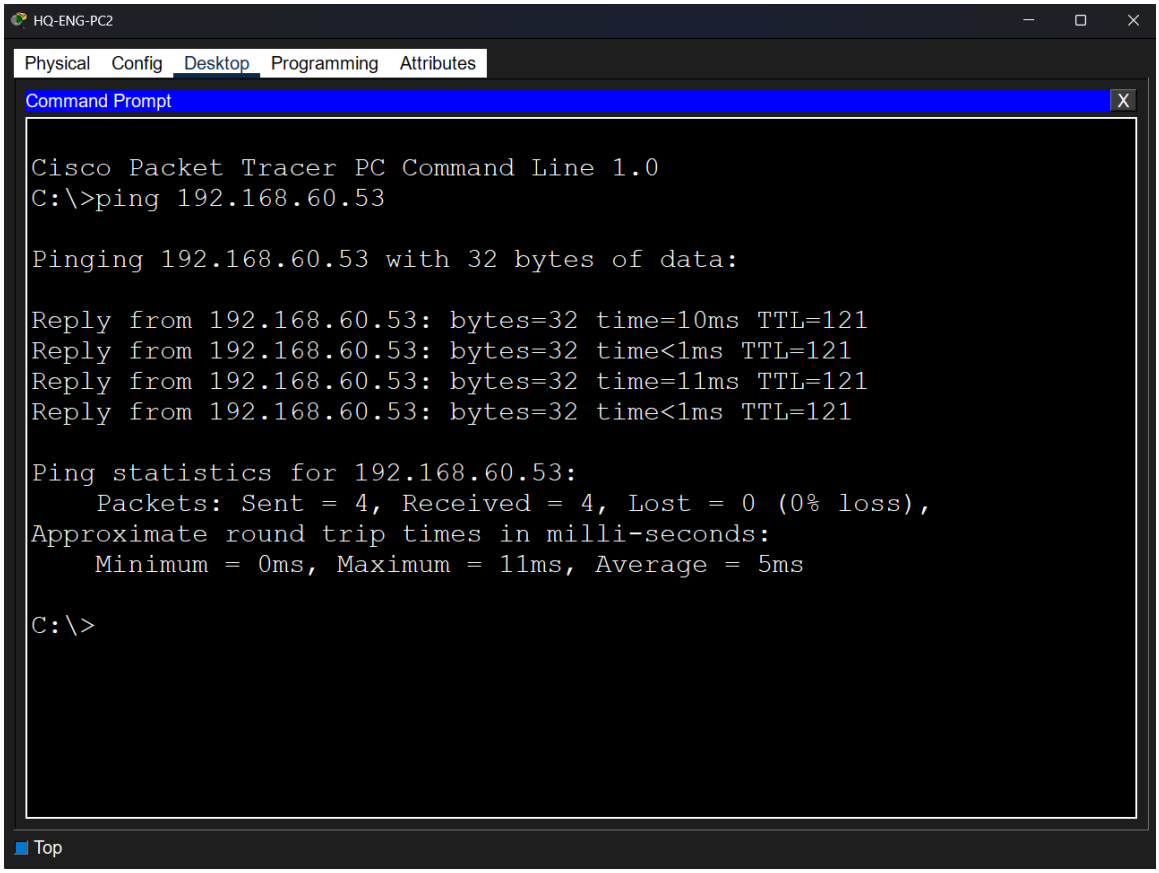
6. Inter-Site Connectivity (HQ ↔ BR)

Description:

This section verifies connectivity between the HQ and BR sites. The successful ping demonstrates that routing between the two sites is correctly configured using dynamic routing protocols.

Screenshot:

This screenshot shows a successful ping from a PC in VLAN 10 (HQ site) to a device in VLAN 60 (BR site) with IP address 192.168.60.53. This confirms end-to-end connectivity between both sites.



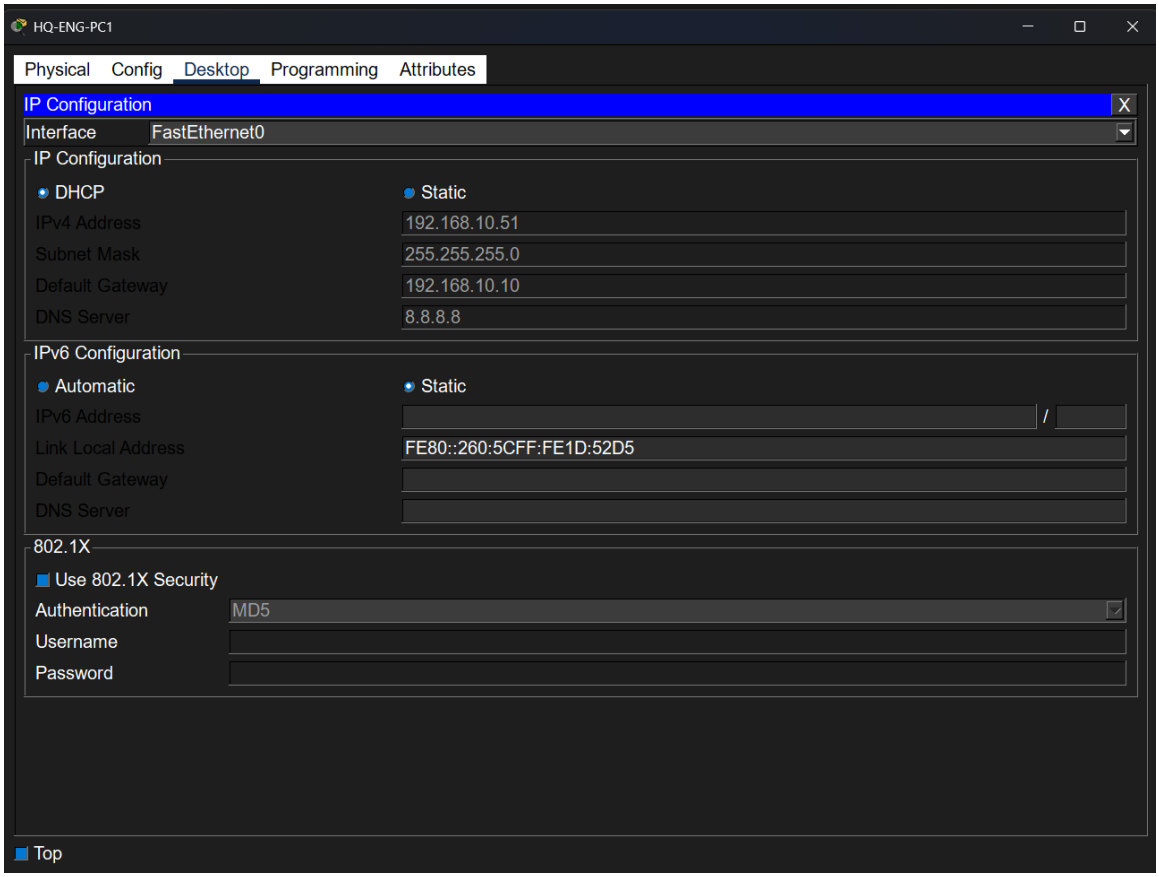
7. DHCP Configuration Verification

Description:

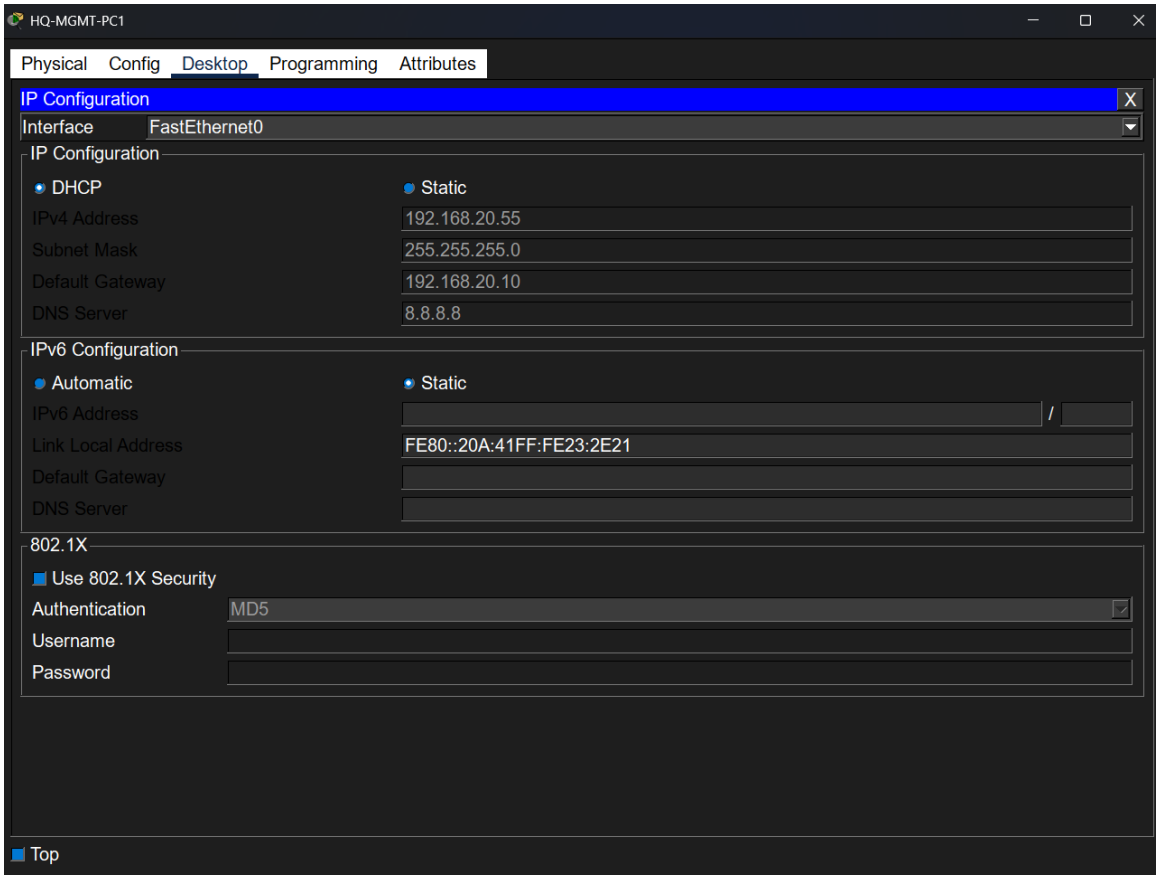
This section verifies that DHCP is correctly configured to automatically assign IP addresses to client devices in all VLANs. Each device receives a valid IP address, subnet mask, and default gateway.

Screenshot:

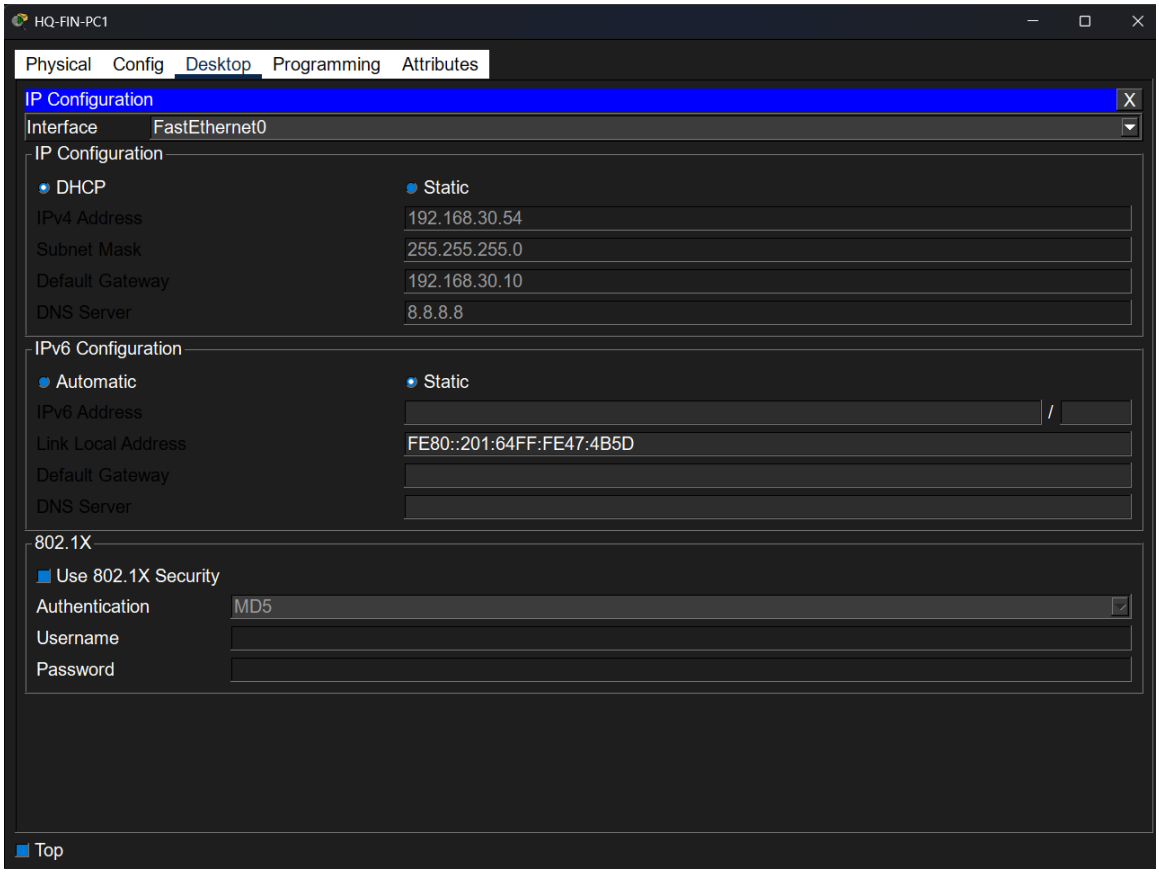
This screenshot shows a PC in VLAN 10 successfully receiving an IP address dynamically from the DHCP server.


HQ-ENG-PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.51
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.10
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:5CFF:FE1D:52D5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

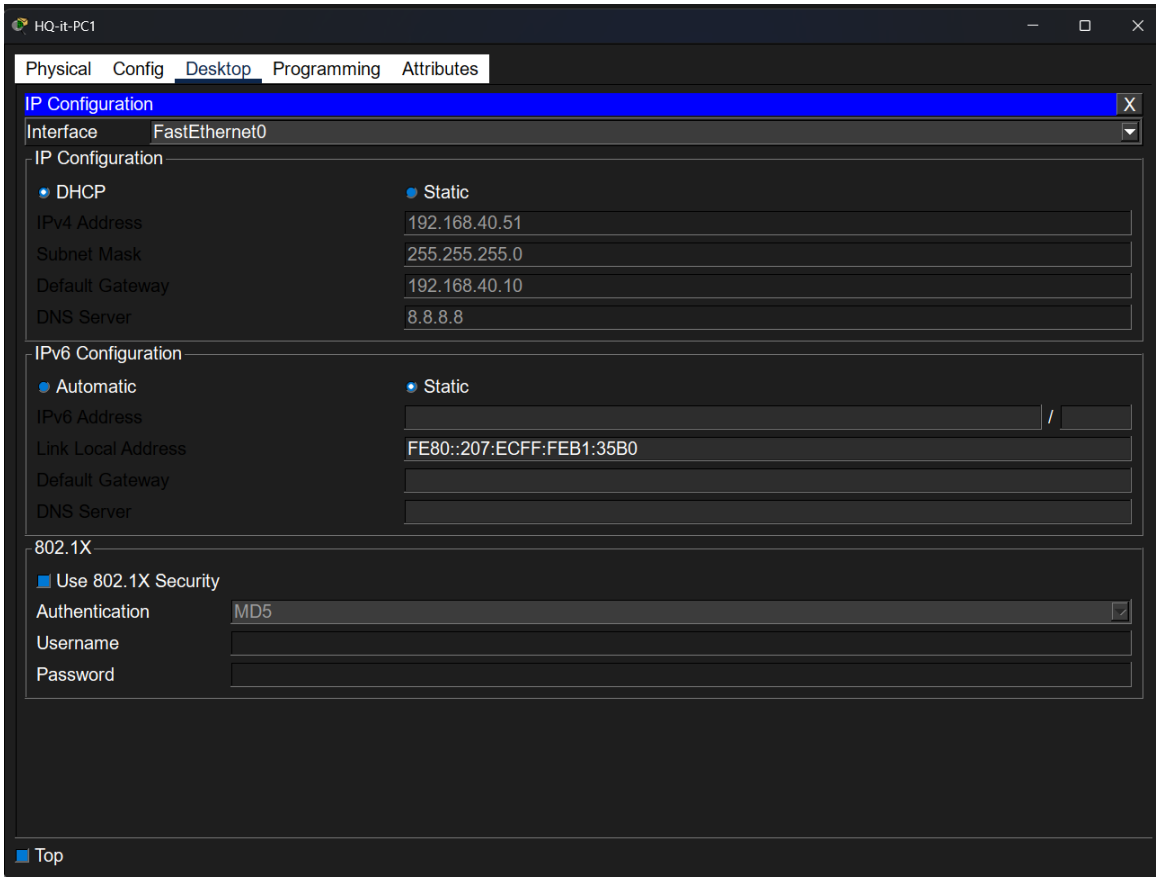
This screenshot shows a PC in VLAN 20 obtaining an IP address via DHCP.



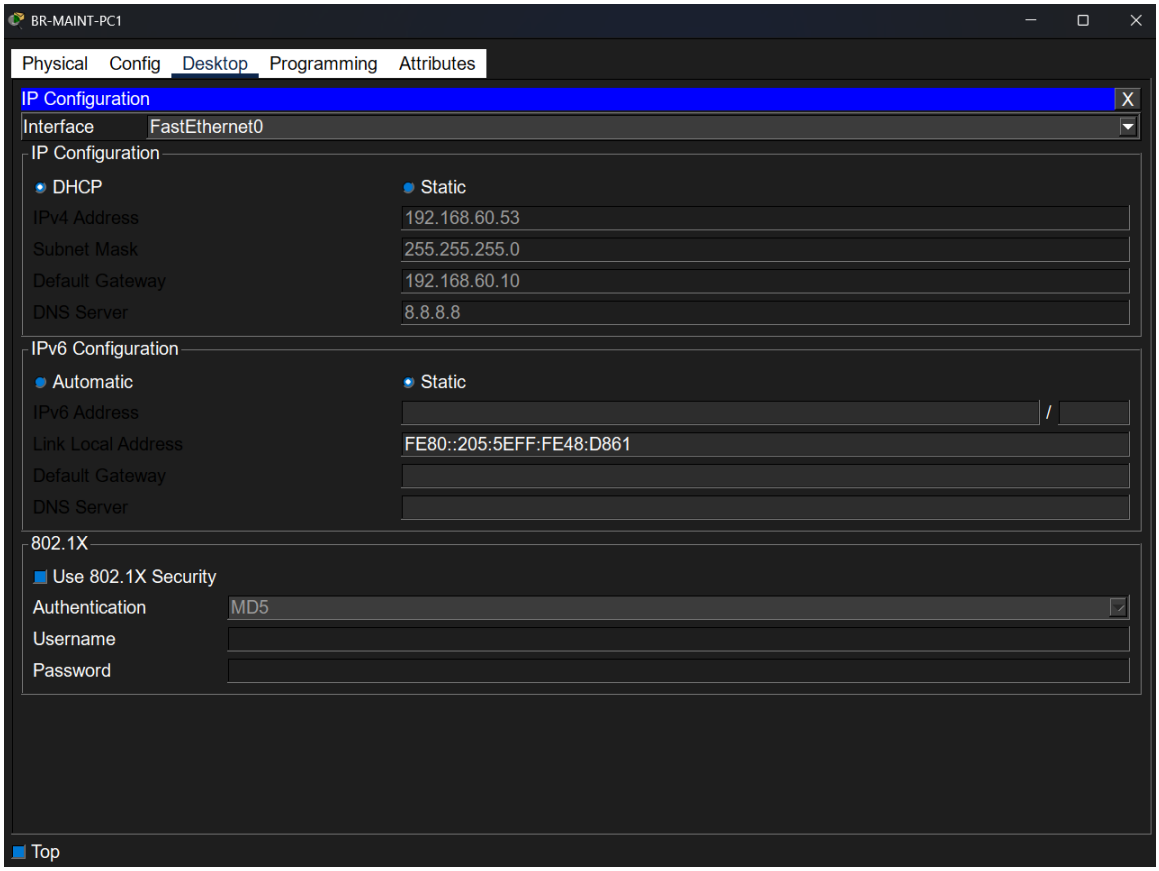
This screenshot shows a PC in VLAN 30 successfully receiving network configuration from DHCP.



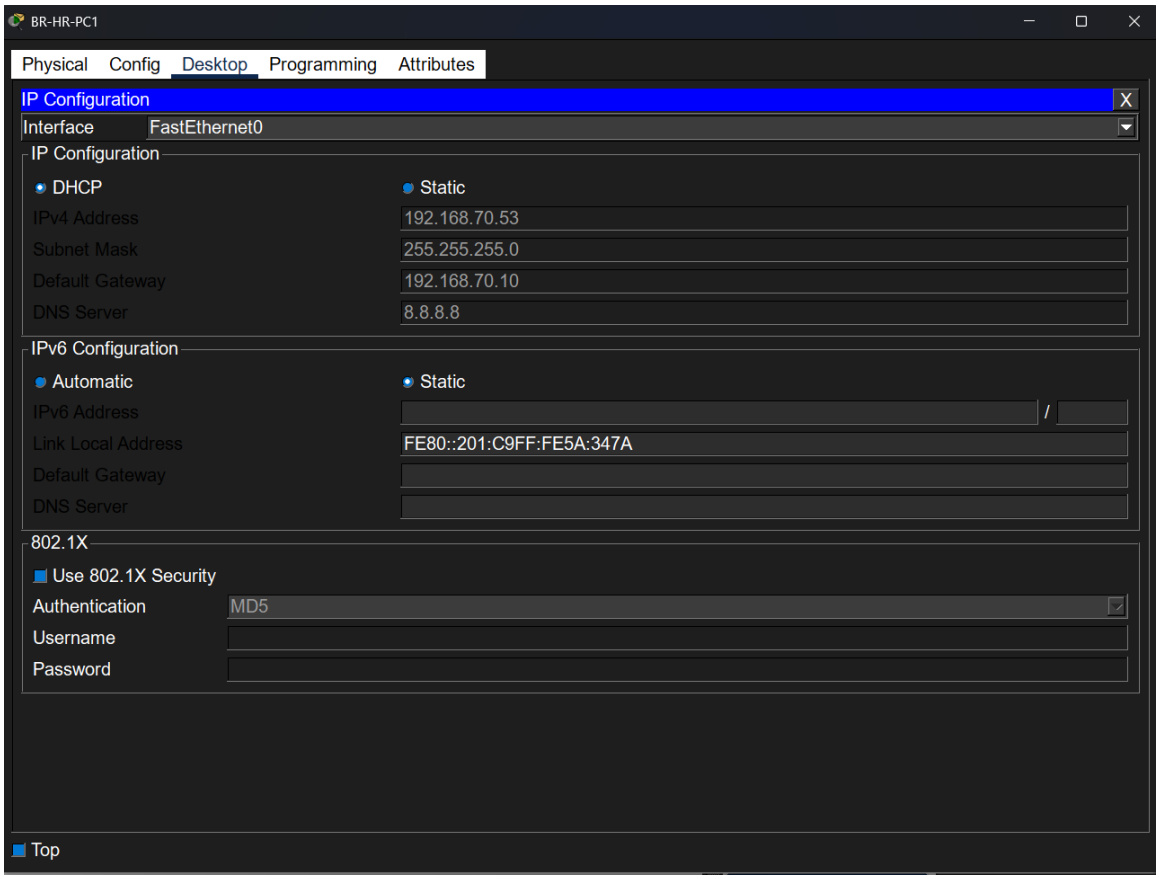
This screenshot shows a PC in VLAN 40 receiving an IP address dynamically.



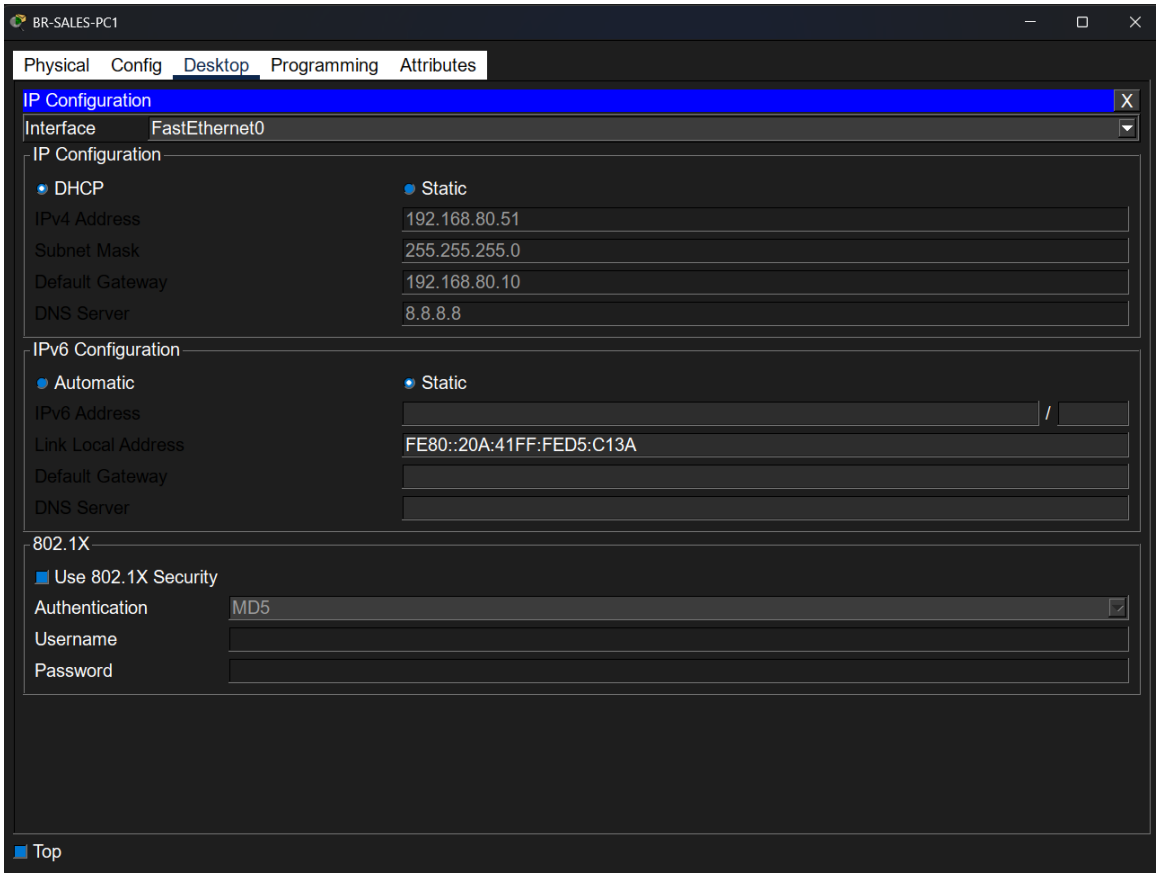
This screenshot shows a PC in VLAN 60 obtaining an IP address from DHCP at the BR site.



This screenshot shows a PC in VLAN 70 successfully receiving an IP address via DHCP.



This screenshot shows a PC in VLAN 80 obtaining a valid IP configuration dynamically.



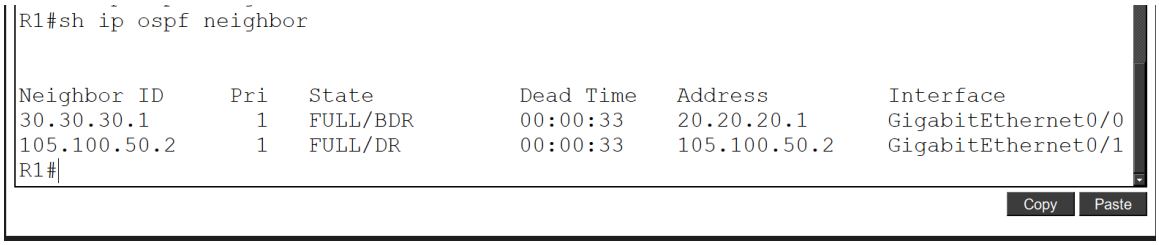
8. OSPF Neighbor Verification

Description:

This section verifies that OSPF is correctly configured between routers. OSPF neighbors are successfully formed, allowing dynamic routing between all networks.

Screenshot:

This section verifies that OSPF is correctly configured between routers. OSPF neighbors are successfully formed, allowing dynamic routing between all networks.



```
R1#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
30.30.30.1        1   FULL/BDR        00:00:33    20.20.20.1      GigabitEthernet0/0
105.100.50.2      1   FULL/DR         00:00:33    105.100.50.2    GigabitEthernet0/1
R1#
```

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This screenshot shows OSPF neighbors on R2, indicating proper routing communication.

```
R2#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
205.200.100.1     1   FULL/DR         00:00:31    30.30.30.2      GigabitEthernet0/0
105.100.50.1      1   FULL/DR         00:00:31    20.20.20.2      GigabitEthernet0/1
R2#
```

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This screenshot shows OSPF neighbor status on R3, confirming that OSPF is functioning correctly.

```
R3#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
205.200.100.2     1   FULL/DR         00:00:36    205.200.100.2   GigabitEthernet0/0
30.30.30.1        1   FULL/BDR        00:00:35    30.30.30.1      GigabitEthernet0/1
R3#
```

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Top

9. HSRP Redundancy Verification

Description:

This section verifies that HSRP is configured to provide gateway redundancy. One switch operates as active and the other as standby for each VLAN.

Screenshot:

This screenshot shows that HQ-DSW1 is operating as the active device for HSRP.

```
HQ-DSW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           192.168.10.2    192.168.10.10
Vl20        20   110 P Active   local           192.168.20.2    192.168.20.10
Vl30        30   110 P Active   local           192.168.30.2    192.168.30.10
Vl40        30   100   Active   local           unknown         192.168.30.10
Vl40        40   110 P Active   local           192.168.40.2    192.168.40.10
HQ-DSW1#
```

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Top

This screenshot shows BR-DSW2 operating as the standby device.

```
HQ-DSW2#sh standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Standby  192.168.10.1    local           192.168.10.10
Vl20        20   100   Standby  192.168.20.1    local           192.168.20.10
Vl30        30   100   Standby  192.168.30.1    local           192.168.30.10
Vl40        40   100   Standby  192.168.40.1    local           192.168.40.10
HQ-DSW2#
```

This screenshot shows BR-DSW1 acting as the active HSRP device.

```
BR-DSW1#sh standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl60        60   110 P Active   local           192.168.60.1    192.168.60.10
Vl70        70   110 P Active   local           192.168.70.2    192.168.70.10
Vl80        80   110 P Active   local           192.168.80.2    192.168.80.10
BR-DSW1#
```

This screenshot shows BR-DSW2 operating as the standby device.

```
BR-DWS2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl60        60   100   Standby  192.168.60.1    local           192.168.60.10
Vl70        70   100   Standby  192.168.70.1    local           192.168.70.10
Vl80        80   100   Standby  192.168.80.1    local           192.168.80.10
BR-DWS2#
```

10. Port Security Verification

Description:

This section verifies that port security is enabled on access ports to restrict unauthorized devices and enhance Layer 2 security.

Screenshot:

This screenshot shows port security enabled on HQ-ASW1 with MAC address learning.

```
HQ-ASW1#sh port-security interface f0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 00E0.8F16.D179:10
Security Violation Count   : 0

HQ-ASW1#
```

This screenshot shows port security configured on BR-ASW1 with appropriate violation mode.

```
BR-ASW1#sh port-security interface f0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0001.43DB.0480:60
Security Violation Count   : 0

BR-ASW1#
```

11. DHCP Snooping Verification

Description:

This section verifies that DHCP snooping is enabled to prevent unauthorized DHCP servers from assigning IP addresses.

Screenshot:

This screenshot shows DHCP snooping enabled on HQ-ASW1.

```
HQ-ASW1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                    Trusted    Rate limit (pps)
-----------------------      -------    ----------------
FastEthernet0/3              no         unlimited
FastEthernet0/4              no         unlimited
FastEthernet0/1              no         unlimited
FastEthernet0/2              no         unlimited
FastEthernet0/5              yes        unlimited
FastEthernet0/6              yes        unlimited
HQ-ASW1#
```

This screenshot shows DHCP snooping configuration on BR-ASW1.

```
BR-ASW1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
60
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                    Trusted    Rate limit (pps)
-----------------------      -------    ----------------
FastEthernet0/1              no         unlimited
FastEthernet0/5              yes        unlimited
FastEthernet0/4              no         unlimited
FastEthernet0/3              no         unlimited
FastEthernet0/2              no         unlimited
FastEthernet0/6              yes        unlimited
BR-ASW1#
```

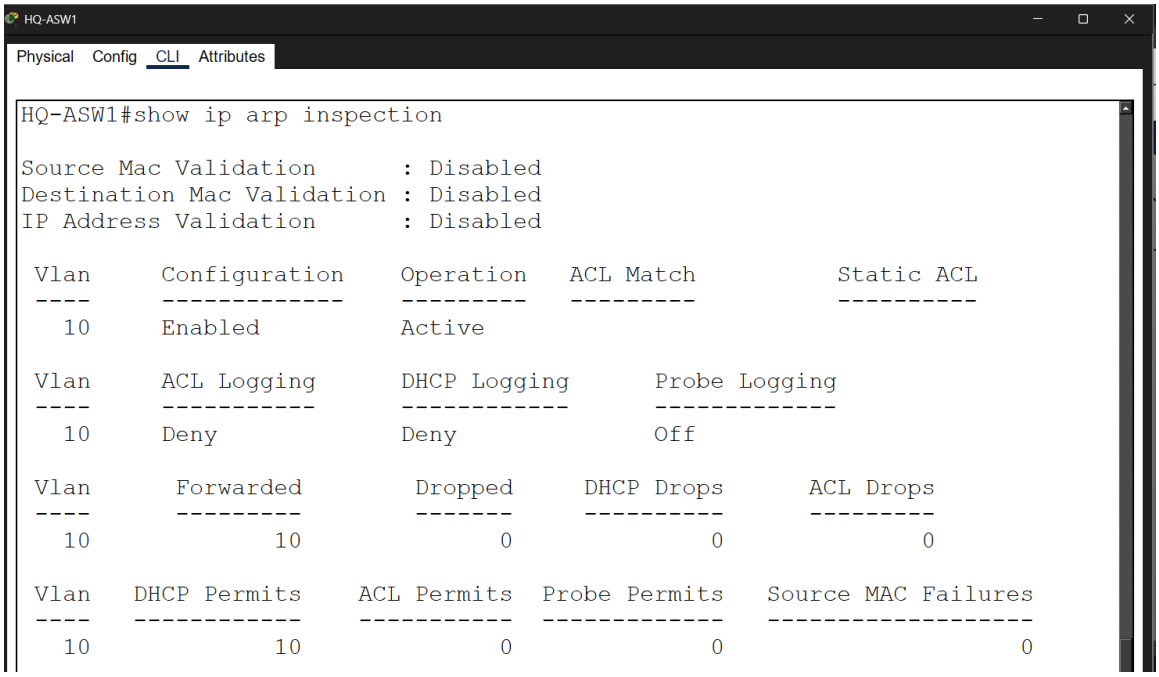
12. Dynamic ARP Inspection (DAI) Verification

Description:

This section verifies that Dynamic ARP Inspection is enabled to protect the network from ARP spoofing attacks.

Screenshot:

This screenshot shows DAI enabled on HQ-ASW1.



This screenshot shows DAI configured and operational on BR-ASW1.

```
BR-ASW1
Physical  Config  CLI  Attributes

BR-ASW1#sh ip arp inspection

Source Mac Validation      : Disabled
Destination Mac Validation : Disabled
IP Address Validation      : Disabled

 Vlan     Configuration    Operation   ACL Match          Static ACL
 ----     -------------    ---------   ---------          ----------
   60     Enabled          Active

 Vlan     ACL Logging      DHCP Logging      Probe Logging
 ----     -----------      ------------      -------------
   60     Deny             Deny              Off

 Vlan      Forwarded        Dropped     DHCP Drops      ACL Drops
 ----      ---------        -------     ----------      ---------
   60              5              0              0              0

 Vlan   DHCP Permits    ACL Permits  Probe Permits   Source MAC Failures
 ----   ------------    -----------  -------------   -------------------
   60              5              0              0                     0
```

13. Access Control List (ACL) Verification

Description:

This section verifies that the ACL is correctly configured on the HQ-Core switch to restrict specific traffic and enhance security.

Screenshot:

This screenshot shows the configured ACL, including denied ICMP and HTTP traffic and permitted other traffic.

```
HQ-CORE#show access-lists
Extended IP access list BLOCK_SERVER
    10 deny icmp any 10.10.10.0 0.0.0.255
    20 deny tcp any 10.10.10.0 0.0.0.255 eq www
    30 permit ip any any (2186 match(es))

HQ-CORE#
```

14. SSH Remote Access Verification

Description:

This section verifies that secure remote access using SSH is configured correctly.

Screenshot:

This screenshot shows a successful SSH connection from a PC in VLAN 30 to HQ-ASW1 using the configured username and password.

```
C:\>ssh -l NetAdmin 192.168.10.3

Password:



HQ-ASW1>
```

Top

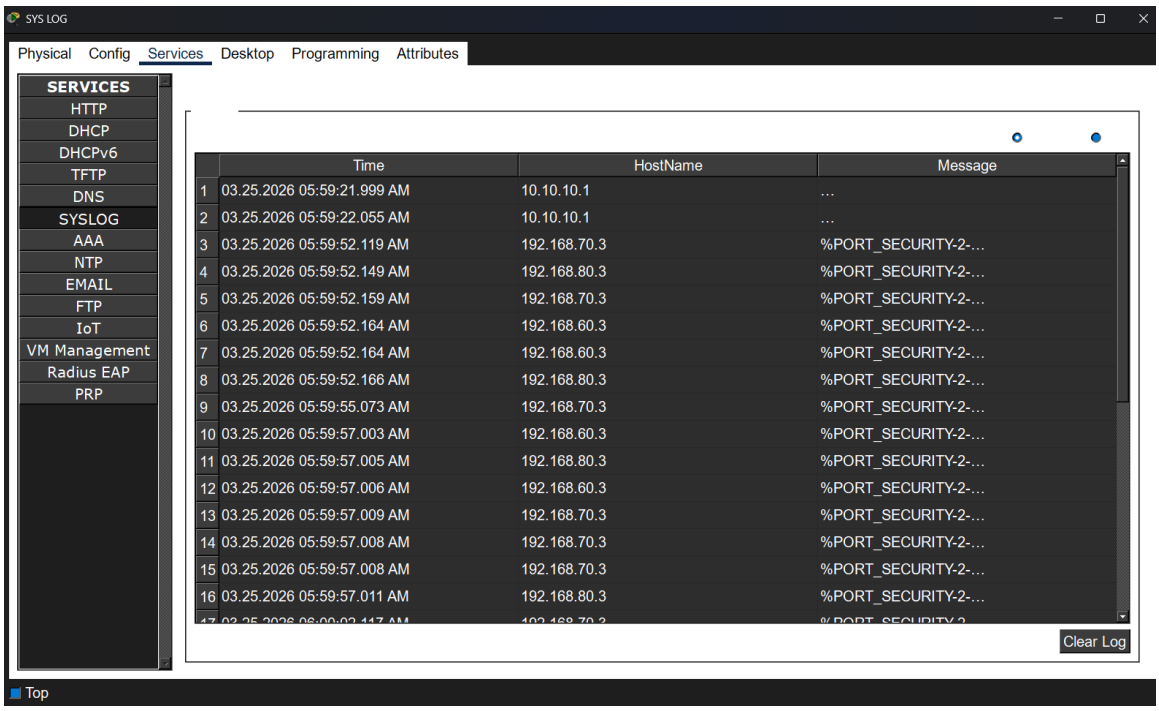
15. Syslog Server Verification

Description:

This screenshot shows log messages received on the Syslog server, confirming successful logging from network devices.

Screenshot:

This screenshot shows log messages received on the Syslog server, confirming successful logging from network devices.



16. NTP Synchronization Verification

Description:

This section verifies that all devices are synchronized with the NTP server to maintain accurate time across the network.

Screenshot:

This screenshot shows that HQ-ASW1 is synchronized with the NTP server.

```
HQ-ASW1#show ntp status
Clock is synchronized, stratum 2, reference is 10.10.10.8
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ED466FD0.000002A1 (6:18:56.673 UTC Wed Mar 25 2026)
clock offset is 6.00 msec, root delay is 10.00  msec
root dispersion is 10.21 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 4, last update was 12 sec ago.
HQ-ASW1#
```

Copy Paste

Top

This screenshot shows that R1 is successfully synchronized.

```
R1#show ntp status
Clock is synchronized, stratum 2, reference is 10.10.10.8
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ED466FE1.00000318 (6:19:13.792 UTC Wed Mar 25 2026)
clock offset is -7.00 msec, root delay is 14.00  msec
root dispersion is 10.23 msec, peer dispersion is 0.24 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 4, last update was 13 sec ago.
R1#
```

Copy Paste

Top

This screenshot shows NTP synchronization on BR-DSW1.

```
BR-DSW1#show ntp status
Clock is synchronized, stratum 2, reference is 10.10.10.8
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ED466FF1.000002AF (6:19:29.687 UTC Wed Mar 25 2026)
clock offset is 8.00 msec, root delay is 21.00  msec
root dispersion is 10.23 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 4, last update was 16 sec ago.
BR-DSW1#
```

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Top

This screenshot shows that BR-ASW1 is correctly synchronized with the NTP server.

```
BR-ASW1#show ntp status
Clock is synchronized, stratum 2, reference is 10.10.10.8
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ED467011.00000304 (6:20:1.772 UTC Wed Mar 25 2026)
clock offset is 12.00 msec, root delay is 10.00  msec
root dispersion is 10.23 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 4, last update was 11 sec ago.
BR-ASW1#
```

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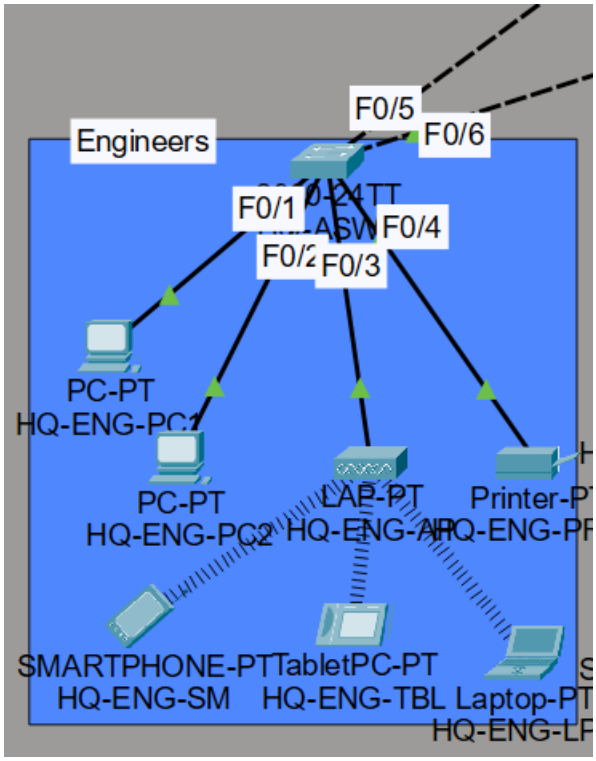
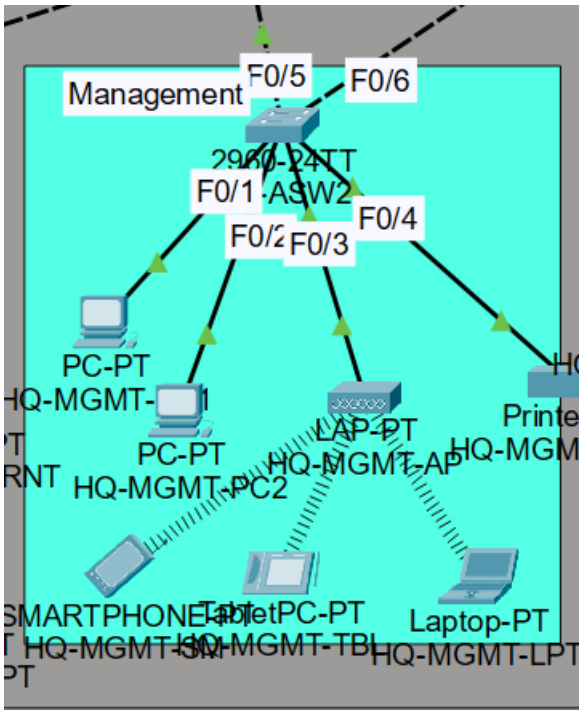
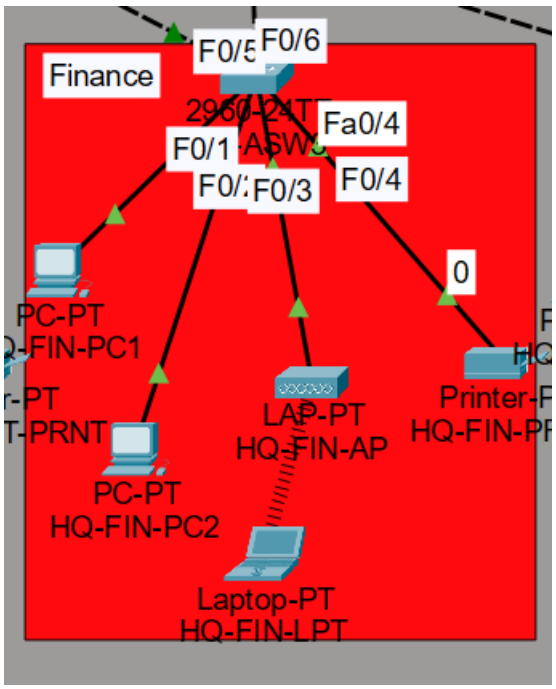
Top

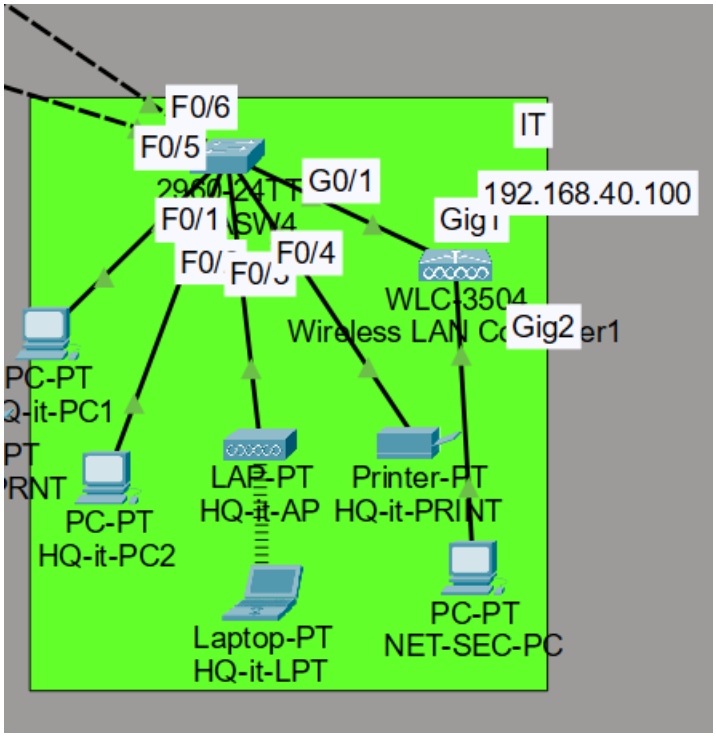
17. Wireless Connectivity Verification

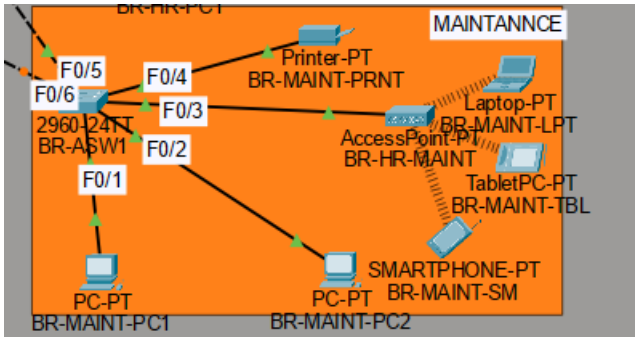
Screenshot:






Finance
F0/6
F0/1
Fa0/4
F0/3
F0/4
0
PC-PT
PC-PT
HQ-FIN-PC2
LAP-PT
HQ-FIN-AP
Laptop-PT


F0/6
F0/5
IT
G0/1
192.168.40.100
F0/1
F0/4
WLC-3504
Gig2
PC-PT
PC-PT
HQ-it-PC2
LAP-PT
HQ-it-AP
Printer-PT
HQ-it-PRINT
Laptop-PT
HQ-it-LPT
PC-PT
NET-SEC-PC


MAINTANNCE
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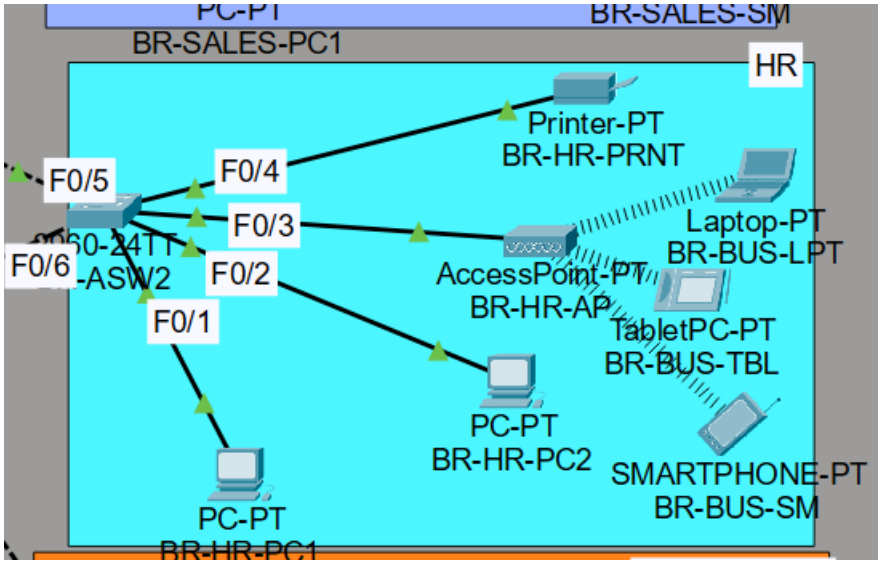
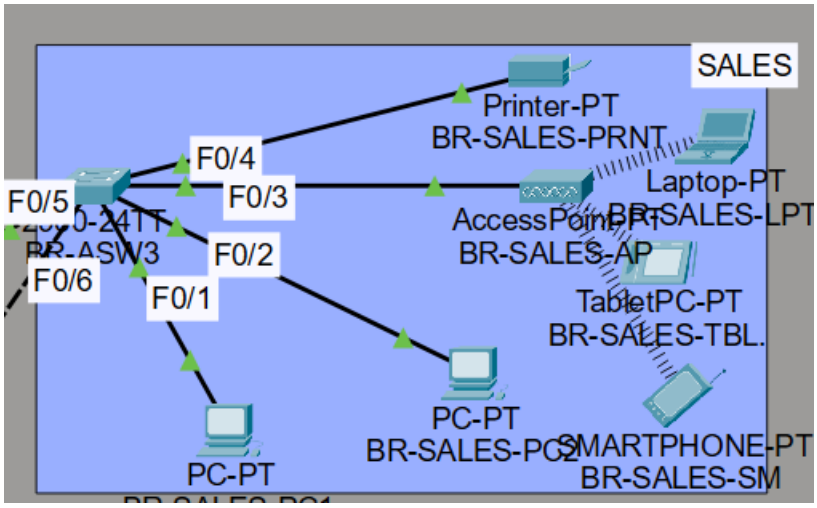
F0/6
F0/4
F0/3
F0/2
F0/1
2960-24TT
BR-ASW1
Printer-PT
BR-MAINT-PRNT
Laptop-PT
BR-MAINT-LPT
AccessPoint-PT
BR-HR-MAINT
TabletPC-PT
BR-MAINT-TBL
SMARTPHONE-PT
BR-MAINT-SM
PC-PT
BR-MAINT-PC1
PC-PT
BR-MAINT-PC2


BR-SALES-PC1
HR
Printer-PT
BR-HR-PRNT
F0/5
F0/4
F0/3
F0/6
F0/2
F0/1
Laptop-PT
BR-BUS-LPT
AccessPoint-PT
BR-HR-AP
TabletPC-PT
BR-BUS-TBL
PC-PT
BR-HR-PC2
SMARTPHONE-PT
BR-BUS-SM
PC-PT


SALES
Printer-PT
BR-SALES-PRNT
F0/4
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F0/3
Laptop-PT
BR-SALES-LPT
F0/2
F0/6
F0/1
TabletPC-PT
PC-PT
BR-SALES-PC2
SMARTPHONE-PT
BR-SALES-SM
PC-PT